314707001 彭甫詔 05/12

15.6

a.

Comparing the OLS estimates for 1987 in column (1) and for 1988 in column (2), the estimated coefficients and their corresponding standard errors are remarkably similar across both years. For instance, the return to experience (EXPER) is 0.1270 in 1987 and 0.1265 in 1988, while the wage penalty for living in the south (SOUTH) shifts slightly from -0.2128 to -0.2384. This close alignment suggests that the cross-sectional relationship between the explanatory variables and log-wages remained relatively stable over this two-year period.

Assumptions Regarding Heterogeneity: When running separate cross-sectional OLS regressions for individual years, it is implicitly assumed that there is **no unobserved individual heterogeneity** across different individuals. Specifically, the regression parameters—including both the intercept and the slopes/marginal effects—are assumed to be **constant and identical for all individuals** in the sample. Any unobserved individual-specific effects (such as innate ability or motivation) are assumed to be non-existent or absorbed entirely into the idiosyncratic error term, under the assumption that they are uncorrelated with the regressors and independent across individuals.

b.

The primary differences in assumptions between the panel data regression model and the separate cross-sectional OLS models in part (a) lie in how they treat individual heterogeneity and error structures over time:

- **Explicit Allowance for Unobserved Heterogeneity:** The cross-sectional OLS models in part (a) assume a constant intercept (beta_1) for all individuals, completely ignoring personal differences. In contrast, model (XR15.6) uses an error components structure, $nu\_\{it\} = u\_i + e\_\{it\}$, where u_i explicitly captures **unobserved individual heterogeneity** that varies across individuals but remains constant over time for each person.

- **Error Structure and Serial Correlation:** The models in part (a) assume that all error terms are independent and identically distributed across all observations. However, because panel data tracks the same individuals over multiple periods, model (XR15.6) recognizes that observations from the same individual are likely correlated over time. Even if the idiosyncratic error e_{it} is serially uncorrelated, the presence of the permanent individual effect u_i introduces a constant covariance between the composite errors of the same individual across different periods ($cov(\nu_{it},\nu_{is}) = \sigma_u^2$).

c.

Comparison between FE and OLS Estimates: When comparing the Fixed Effects (FE) estimates in column (3) with the cross-sectional OLS estimates in columns (1) and (2), we observe noticeable changes in all coefficients after controlling for unobserved individual heterogeneity (u_i).

- The coefficient on EXPER drops substantially from around 0.1270 in OLS to 0.0575 in FE.
- The coefficient on UNION decreases slightly from 0.1445 (1987) / 0.1102 (1988) to 0.0822.
- The coefficient on SOUTH becomes more severely negative, moving from -0.2128 (1987) / -0.2384 (1988) to -0.3261.

Coefficient Showing the Most Difference:
Apart from the intercept, the coefficient for **SOUTH** shows the largest absolute difference. Its value changes from an average of approximately -0.2254 in the OLS regressions to **-0.3261** in the FE model, representing an increase in the negative wage penalty for living in the South by roughly 0.10.

d.

Degrees of Freedom: According to the formula for the F-test testing for individual differences, the degrees of freedom are defined as:

Numerator(df_1)= N - 1

Denominator (df_2) = NT - N - K_s

Given from the sample data: $N = 716$, $T = 2$, $NT = 1432$, and $K_s = 4$

- Numerator(df_1)= 716 - 1 = 715
- Denominator (df_2) = 1432 - 716 - 4 = 712

Therefore, if the null hypothesis is true, the test statistic follows an F-distribution with (715, 712) degrees of freedom.

Critical Value at the 1% Significance Level:

Using the exact degrees of freedom $F(715, 712)$, the critical value at the 1% level of significance ($\alpha = 0.01$) obtained via statistical software is approximately **1.134**.

Conclusion:

Since the calculated F-statistic of **11.68** is substantially larger than the 1% critical value of **1.134**, we **strongly reject the null hypothesis** of no individual differences. We conclude that there is highly significant unobserved individual heterogeneity among the young women in the sample. This implies that a pooled OLS model is misspecified, and controlling for individual fixed effects is statistically necessary.

e.

Different Assumptions Implemented:

- **Without Cluster-Robust SEs (Column 3):** This model relies on the classical Gauss-Markov assumptions for panel data, which assume that the idiosyncratic errors (e_{it}) are **conditionally homoskedastic** and **serially uncorrelated** over time within each individual.
- **With Cluster-Robust SEs (Column 4):** This approach relaxes those rigid assumptions by allowing the idiosyncratic errors (e_{it}) within the same cluster (the same individual young woman) to exhibit arbitrary forms of **heteroskedasticity** and **serial correlation** over time. It only strictly requires that the errors are uncorrelated *across* different individuals.

Comparison of Standard Errors:

Comparing the standard errors in column (4) with those in column (3), we see that the robust standard errors are generally **larger** than the conventional ones. The variable that shows the most substantial and dramatic difference is **SOUTH**. Its standard error nearly doubles, expanding from **0.1258** in column (3) to **0.2495** in column (4). The noticeable inflation of this standard error implies the presence of strong within-individual serial correlation in the error components over time. Ignoring this correlation (as in column 3) would lead to overstating the precision of our coefficients and obtaining artificially inflated t-statistics.

f.

Coefficient Showing the Most Difference:

Apart from the intercept, the coefficient for **SOUTH** shows the largest absolute difference between the FE model in column (3) and the RE model in column (5). Its value is **-0.3261** under FE and shifts to **-0.2326** under RE, resulting in an absolute discrepancy of **0.0935**. The coefficient for $EXPER$ also shows a meaningful difference, moving from 0.0575 (FE) to 0.0986 (RE).

Hausman Test Calculation and Comparison Column Choice:

Using the Hausman test statistic formula (15.36) for the individual coefficient of EXPER:

$$t = \frac{b_{FE} - b_{RE}}{\sqrt{se(b_{FE})^2 - se(b_{RE})^2}} = \frac{0.0575 - 0.0986}{\sqrt{0.0330^2 - 0.0220^2}} = \frac{-0.0411}{0.0246} \approx -1.$$

- The classical Hausman test relies strictly on the assumption that under the null hypothesis (no endogeneity), the random effects (FGLS) estimator is **asymptotically efficient** (achieving the lowest possible variance). If we utilize the cluster-robust standard errors from column (4), the efficiency requirement is violated due to heteroskedasticity or serial correlation, rendering the standard Hausman denominator formula invalid and theoretically incorrect. Thus, the test must be performed using conventional standard errors from column (3).

Appropriateness of the Random Effects Model: Based on the systematic differences observed between the FE and RE coefficients, there is clear evidence that the unobserved individual heterogeneity (u_i) is correlated with the explanatory variables. This indicates the presence of severe endogeneity. Under endogeneity, the Random Effects estimator is **biased and inconsistent**, whereas the Fixed Effects estimator remains **consistent**. Therefore, the random effects estimation in this model is **not appropriate**, and the Fixed Effects model should be selected instead.

15.20

a.

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 437.76425    1.34622 325.180  < 2e-16 ***
small         5.82282    0.98933   5.886 4.19e-09 ***
aide          0.81784    0.95299   0.858    0.391    
tchexper      0.49247    0.06956   7.080 1.61e-12 ***
boy          -6.15642    0.79613  -7.733 1.23e-14 ***
white_asian   3.90581    0.95361   4.096 4.26e-05 ***
freelunch   -14.77134    0.89025 -16.592  < 2e-16 ***
```

Small Classes (SMALL): Yes, students perform significantly better. Being in a small class increases the reading score by **5.82 points** ($p < 0.001$).

Teacher's Aide (AIDE): No, having an aide does not statistically improve scores. The effect is small (**0.82 points**) and highly insignificant ($p = 0.391$).

Teacher's Experience ($TCHEXPER$): Yes, each additional year of teacher experience significantly raises scores by **0.49 points** ($p < 0.001$).

Sex/Race (BOY / $WHITE\_ASIAN$): Yes, both matter. Boys score **6.16 points lower** than girls ($p < 0.001$), while White/Asian students score **3.91 points higher** ($p < 0.001$).

b.

School Fixed Effects (Within) Equation:

$$\begin{aligned} widehat\{readscore\}\_\{it\} &= 6.4902\, small\_\{it\} + 0.9961\, aide\_\{it\} + 0.2856\, tchexper\_\{it\} \\ &- 5.4559\, boy\_\{it\} + 8.0280\, white\backslash\_asian\_\{it\} \\ &- 14.5936\, freelunch\_\{it\} \end{aligned}$$

No, none of the primary conclusions have changed. The effect of small classes (small) remains positive and highly significant, shifting slightly from **5.82** to **6.49** ($p < 0.001$).

- The effect of a teacher's aide (aide) remains small (**0.99**) and statistically insignificant ($p = 0.2586$).
- The coefficients are remarkably stable because the STAR experiment is a **Randomized Controlled Trial (RCT)**. Since students and teachers were randomly assigned within schools, the classroom traits are independent of the unobserved school characteristics, meaning that adding school fixed effects does not alter the core conclusions.

c.

Test Result:

$$F = 16.698, \quad df_1 = 78, \quad df_2 = 5681, \quad p\text{-value} < 2.2 \times 10^{-16}$$

Since the p-value is virtually zero, we **strongly reject the null hypothesis** of no individual effects. This indicates that school fixed effects are highly statistically significant.

Condition for Minimal Influence

The inclusion of significant fixed effects will have little influence on other coefficients if **the unobserved school heterogeneity (u_i) is completely uncorrelated to the explanatory variables** (cov(x_{kit}, u_i) = 0). In this sample, this condition is satisfied due to the **Randomized Controlled Trial (RCT)** design. Because students and teachers were randomly assigned within schools, class-type traits (like small and aide) are independent of unobserved school characteristics. Therefore, adding school fixed effects does not introduce omitted variable bias to these randomized coefficients, keeping their estimates stable.

d.

Comparison of Estimates

The Random Effects (RE) coefficients are very close to the Fixed Effects (FE) estimates from part (b). For example, the effect of small is **6.46** in RE, compared to **6.49** in FE and **5.82** in OLS. This stability confirms that the primary treatment effects are heavily anchored.

Correlation with School Effects:

Since the STAR experiment is a randomized controlled trial, the classroom traits (small, aide) and student traits (boy, white_asian, freelunch) were randomly assigned within schools and are uncorrelated with the school effects (u_i). However, **tchexper (teacher's experience)** is a variable that might be correlated with school effects, as higher-quality or wealthier schools (u_i) often have more resources to attract or retain senior teachers. Since the p-value is virtually zero, we **strongly reject the null hypothesis** of no random effects (H_0: sigma_u^2 = 0). There is decisive empirical evidence of significant unobserved school-level heterogeneity, meaning that a standard pooled OLS model is inefficient and inappropriate.

e.

Hausman t-test Results (with 5% Critical Value ± 1.96):

- `small`: $t = \mathbf{1.146}$ (Statistically insignificant)
- `aide`: $t = \mathbf{0.128}$ (Statistically insignificant)
- `tchexper`: $t = -\mathbf{1.938}$ (Borderline insignificant, $|-1.938| < 1.96$)
- `white_asian`: $t = \mathbf{1.218}$ (Statistically insignificant)
- `freelunch`: $t = -\mathbf{0.096}$ (Statistically insignificant)

Implications

Since all calculated Hausman t-statistics have an absolute value **less than 1.96**, we **fail to reject the null hypothesis** (H_0: cov(x_{kit}, u_i) = 0) for each individual coefficient. This implies that there is no strong evidence of individual-level endogeneity for these variables. The random effects model appears to provide consistent estimates for these parameters.

What Happens to boy (Why NaN occurs)

When applying the test to boy, R returns **NaN**. This happens because the estimated variance of the Random Effects estimator for boy is slightly larger than that of the Fixed Effects estimator (se(b_{RE})^2 > se(b_{FE})^2). As a result, the term under the square root in the denominator (se(b_{FE})^2 - se(b_{RE})^2) becomes **negative**, making it mathematically impossible to calculate a real t-statistic. This illustrates a classic small-sample limitation of the traditional Hausman test.

f.

Mundlak Test Results:
We conducted a joint Wald test on the coefficients of all the added school-average variables ($H_0 : \gamma = 0$).

- χ^2 **Statistic:** `13.683`
- p**-value:** `0.03338`

Since the p-value is significant (less than 0.05), we **reject the null hypothesis**. This indicates that the school-averages are jointly statistically significant. Econometrically, this confirms that the unobserved school heterogeneity (u_i) is indeed correlated with one or more of the explanatory variables. Consequently, the Random Effects estimator is inconsistent, and the **Fixed Effects model should be preferred** for valid causal inference.